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ORIGINAL RESEARCH

Protocol Standardization May Improve Precision Error of InBody 720 Body Composition Analysis

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Background: Bioelectrical impedance analysis (BIA) is a popular technique which can be used to track longitudinal changes in body composition. However, precision of the technique has been questioned, especially among athletic populations where small but meaningful changes are often observed. Guidelines exist which attempt to optimize precision of the technique but fail to account for potentially important variables. Standardization of dietary intake and physical activity in the 24 hr prior to assessment has been proposed as an approach to minimizing the error of impedance-derived estimates of body composition. ***Methods:*** Eighteen recreational athletes, male ($n = 10$) and female ($n = 8$), underwent two consecutive BIA tests to quantify within-day error, and a third test (the day before or after) to quantify between-day error. All food and fluid intake plus physical activity from the 24 hr prior to the first BIA scan was replicated during the following 24 hr. Precision error was calculated as the root mean square standard deviation, percentage coefficient of variation, and least significant change. ***Results:*** There were no significant differences in precision error of within- and between-day fat-free mass, fat mass, and total body water. Differences in precision error of fat-free mass and total body water, but not fat mass, were less than the smallest effect size of interest. ***Conclusion:*** The 24-hr standardization of dietary intake and physical activity may be an effective approach to minimizing precision error associated with BIA. However, further research to confirm the validity of this protocol compared to nonstandardized or randomized intake is warranted.

Keywords: bioelectrical impedance analysis, bioelectrical impedance spectroscopy, least significant change, athletes

Body composition is one of an array of factors that has been found to influence athlete performance across a broad range of sports (Slater et al., 2017). As such, it has become common practice for sports science practitioners to routinely monitor changes in athlete body composition over time, allowing them to assess the efficacy of nutrition and training interventions (Meyer et al., 2013; Slater et al., 2017). However, athletes typically experience relatively small changes in body composition over time, and therefore, techniques used to assess body composition must have high precision (Moon, 2013).

Bioelectrical impedance analysis (BIA) is a popular technique that allows body composition to be quantified in a cheap, safe, and noninvasive way (Kerr & Hume, 2018). The BIA technique works by introducing an electrical current into the body at known frequencies and measuring the impedance to current flow, which varies between tissues depending on fluid and electrolyte content. Body composition is predicted based on the assumption of a constant hydration fraction (73%) of the fat-free mass (FFM; Kyle et al., 2004b). In recent years, the reliability of impedance for assessing athlete body composition longitudinally has been questioned (Farley et al., 2020), in part due to the plethora of factors (Brantlov et al., 2017; Kushner et al., 1996) which can impact on precision error. Such factors can be categorized as either technical (i.e., pertaining to the machine or technician), for example, postural position and electrical interference; or biological (i.e., pertaining to variability of subject presentation), for example, diet, exercise, and hydration status (Kushner et al., 1996). Farley et al. (2020) found that between-day least significant change (LSC) was 3,966 g and 3,607 g FFM and fat mass (FM) respectively, indicating that large changes in body composition would need to occur in order to be distinguishable from the error of the technique.

To enhance the precision of impedance measurements, a set of best practice guidelines has been developed (Kyle et al., 2004a) that attempts to minimize both the technical and biological errors associated the impedance technique. While such guidance is likely of value, it fails to account for between-day variability in dietary intake and physical activity, which have been shown to impact intracellular solute content, including glycogen (Bone et al., 2017; Shiose et al., 2018) and electrolytes (Costill et al., 1976), for 24 hr and beyond. The current approach of an overnight fast has been shown to be insufficient to return BIA estimates to baseline following acute dietary manipulations (Tinsley et al., 2017). This is likely due to such shifts in muscle solute content caused by day-to-day variance in diet and physical activity.

Recently, the 24-hr replication of dietary intake and physical activity has been proposed as a new approach to enhancing the precision error of between-day estimates of body composition (Farley et al., 2020). Indeed, highly precise measurements have been observed when dietary intake is standardized in the 24 hr prior to testing (Rose et al., 2021). However, the impact of this approach on the precision error of impedance estimates has not been explored. Given both variance in hydration status (Thompson

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et al., 1991) and muscle solute content (Shiose et al., 2018) have been shown to significantly influence impedance estimates, this warrants further exploration. Therefore, the aim of this investigation was to assess the impact of 24-hr diet and physical activity standardization on consecutive-day precision.

Methods

Sample

A convenience sample of healthy males and females between the ages of 18 and 40 years was recruited to participate in this study. To be eligible, participants could not take medications that interfere with BIA measurements and female participants were required to confirm they were not pregnant. A total of ten male and eight female recreational athletes were recruited to participate in this study (Table 1). Study participants met the Tier 1 definition of "recreationally active" as described in McKay et al. (2022) and included individuals from a variety of different sports. All participants were informed of the possible risks and benefits of the research and provided their written consent to partake in the study. Ethics approval was obtained from the Regional Committees for Medical and Health Research Ethics in Norway (2017/2160).

Experimental Design

A test–retest study design was employed to quantify the precision error of within- and between-day estimates of body composition. All participants undertook a total of three identical BIA measurements across a 48-hr testing period, including two duplicate within-day tests (WDT1 and WDT2) to quantify technical error, and a third, between-day test (BDT1) to quantify biological error, which was randomly assigned on the day before or after the repeat tests. A computer was used to generate the random allocation sequence. Measurements of weight, FFM, FM, and total body water (TBW) were recorded at each test.

Participant Preparation

On the morning of each test day, all participants presented in overnight (>8 hr) fasted and rested (no physical activity) states as per the best practice guidelines outlined by Kyle et al. (2004a). This included participants wearing underwear with no metal components (e.g., zippers and studs) and removing all jewelry. Participants also voided their bladder prior to testing.

To achieve standardization of diet and exercise, guidance was provided to participants to replicate all dietary intake and physical activity for a 24-hr period. Participants were instructed to consume their normal diet and partake in recreational activity as desired in the day prior to their first test. To support presentation in a euhydrated state, participants were encouraged to drink one to two glasses of water with each meal or snack. It was recommended they select enjoyable, easily accessible foods and only partake in training that could be easily repeated on consecutive days. Participants recorded all food and fluids consumed and physical activity undertaken (type, time, duration, and intensity) across a 48-hr period in a standard diet and training diary to confirm replication of diet and training.

Bioelectrical Impedance Analysis

Body mass and stature were measured at the start of each testing session using balance scales and a stadiometer with participants wearing underwear. The specific gravity of the first morning urine was measured using a digital refractometer (UG-Alpha, Atago Corporation). Body composition was then measured using the InBody 720 analyzer (InBody). The BIA device was calibrated as per the manufacturer's instructions, with participant stature, body mass, age, and sex programmed into the unit. Participants were then instructed to stand on the machine scale and grip the handles with arms held out to their sides at an approximate 15° angle, ensuring that the palms, fingers, and soles of the feet were in contact with the electrodes. They were asked to remain still during measurements while electrical currents were administered at frequencies of 1, 5, 50, 250, 500, and 1,000 kHz to measure impedance. WDTs were undertaken immediately after each other, with volunteers remaining on the InBody 720 between measurements.

Data Analysis

FoodWorks (version 10.0, Xyris Software) was used to quantify participants' total food and fluid intake for each day, including a breakdown of macro- and micronutrient intake. Analysis of the body composition data was performed in Microsoft Excel and

Table 1 Summary of Participant Characteristics

	Cohort (n = 18)	Females (n = 8)	Males (n = 10)
Age (years)	31 ± 8	30 ± 10	32 ± 6
WDT1 mass (kg)	75.7 ± 14.6	62.6 ± 8.1	86.1 ± 8.8
WDT2 mass (kg)	75.6 ± 14.6	62.6 ± 8.1	86.1 ± 8.8
BDT1 mass (kg)	75.7 ± 14.5	62.7 ± 8.0	86.1 ± 8.8
Average mass (kg)	75.7 ± 14.6	62.6 ± 8.0	86.1 ± 8.8
Height (cm)	177.1 ± 9.1	170.66 ± 8.16	182.18 ± 6.23
BMI (kg/m^2)	24.0 ± 3.5	21.4 ± 2.0	25.9 ± 3.0
USG	1.021 ± 0.006	1.017 ± 0.007	1.024 ± 0.003
FFM (g)	62,617 ± 12,544	64,017 ± 9,386	63,127 ± 12,058
FM (g)	12,457 ± 6,494	13,246 ± 5,638	13,313 ± 5,508
TBW (ml)	45,876 ± 9,241	46,883 ± 6,877	46,257 ± 8,884

Note. WDT1 = within-day test 1; WDT2 = within-day test 2; BDT1 = between-day test 1; BMI = body mass index; USG = urine specific gravity; FFM = fat-free mass; FM = fat mass; TBW = total body water.

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RStudio Desktop (version 1.4.1717; RStudio, 2021). Both the within- and between-day test–retest differences were calculated from the raw FFM, FM, and TBW measurements, as generated by the InBody 720 device. Paired *t* tests were used to test for significant differences using an alpha value of .025 (adjusted downward for repeated tests) and Pearson's correlation coefficients were used to test the strength of correlations. Bland–Altman plots were developed to assess the data for systematic bias using the "ggplot2" package in RStudio.

Precision was calculated as the root mean square standard deviation (RMS_*SD*) and percentage coefficient of variation (%CV). Paired *t* tests were performed to test for significant differences in the within- and between-day %CV with the alpha value set at .05. LSC values were calculated by multiplying the RMS_*SD* by 2.77, which is the standard adjusting factor derived from *z* score that corresponds to a 95% level of confidence. Equivalence testing with the TOSTER package in RStudio was used to determine whether the 95% confidence intervals (CIs) for the mean differences in %CV were greater than the smallest effect size of interest. The upper equivalence bound was set at 1% to correspond with previously described levels of interrater reliability for anthropometric techniques (Kerr et al., 2017; Rose et al., 2021). If the 95% confidence limit for the difference between within- and between-day %CV exceeded the upper confidence limit, then the null hypothesis that $\Delta \geq 1\%$ was rejected.

Results

The diet and training diaries confirmed successful replication of dietary intake across the 2-day period, as evident through no significant ($p > .05$) differences in participants' intake of total energy (9,606 ± 2,311 kJ vs. 9,616 ± 2,304 kJ, $p = .77$), carbohydrate (283 ± 82 g vs. 283 ± 81 g, $p = .89$), fat (73 ± 14 g vs. 73 ± 15 g, $p = .67$), protein (124 ± 34 g vs. 124 ± 35 g, $p = .64$), fluid (1,986 ± 317 ml vs. 2,001 ± 326 ml, $p = .26$), or sodium intake (1,881 ± 283 mg vs. 1,890 ± 326 mg, $p = .52$). Replication of physical activity was also confirmed through participants' self-reported training activities as recorded in the diet and training diaries. There was no significant difference in participant urine specific gravity on Day 1 compared to Day 2 of testing ($p = .47$). There were also no significant difference in body mass (in kilograms) measured on Days 1 and 2 of testing ($p = .84$).

There were no significant differences in within-day FFM ($p = .48$), FM ($p = .15$), or TBW ($p = .52$) measurements. There was also no significant differences in between-day FFM ($p = .75$), FM ($p = .80$), or TBW ($p = .76$) measurements. The Pearson correlation coefficients showed near-perfect correlations for the plots of the WDTs (WDT1 and WDT2) for FFM ($R^2 = .9981$), FM ($R^2 = .9929$), and TBW ($R^2 = .9983$; Figure 1). There were also near-perfect correlations for plots of the BDTs (WDT1 and BDT1) for FFM ($R^2 = .9978$), FM ($R^2 = 9933$), and TBW ($R^2 = .9979$; Figure 1). Bland–Altman plots showing agreement of within- and between-day measurements are presented in Figure 2.

Within-day precision for FFM, FM, and TBW is presented in Table 2, including RMS_*SD*, %CV, and LSC. Similarly, between-day precision is presented in Table 3. Between-day RMS_*SD* and LSC were higher than within-day values for FFM and TBW but were lower than the within-day values for FM. No significant differences in within- and between-day %CV for FFM ($p = .36$), FM ($p = .82$), or TBW ($p = .20$) were found.

Figure 3 presents the equivalence tests which show whether the difference in within- and between-day %CV was less than or equal to the smallest effect size of interest ($\Delta \geq 1\%$). The difference in within- and between-day %CV was statistically equivalent to 0 for FFM ($\Delta = 0.14$, 95% CI [−0.013, 0.29]) and TBW ($\Delta = 0.17$, 95% CI [−0.14, 0.48]), with the 95% CIs exceeding the upper equivalence bound, indicating that the difference in %CV was not greater than the smallest effect size of interest of 1%. However, the difference in within- and between-day %CV was not statistically equivalent to 0 for FM ($\Delta = 0.20$, 95% CI [−1.71, 2.11]), with the 95% CI exceeding the upper equivalence bound indicating that $\Delta \geq 1\%$.

Discussion

BIA has gained popularity as a method of measuring body composition due to the accessibility and affordability of the technique (Meyer et al., 2013). However, the reliability of BIA has come into question due to the wide range of technical and biological factors that have the potential to influence measurement precision (Farley et al., 2020; Rose et al., 2021). A set of standard best practice guidelines has been developed to minimize the impact of these factors (Kyle et al., 2004a); however, these guidelines do not account for day-to-day fluctuations in the hydration status and muscle solute content caused by differences in client presentation, specifically, variations in dietary intake and physical activity. The standardization of diet and physical activity modeled in this study has the potential to enhance the precision of BIA estimates of FFM, FM, and TBW in a recreationally active population. This is evidenced by the results of the study which showed no significant difference ($p > .025$) in between-day FFM, FM, or TBW and small LSC (BDT1 LSC; FFM = 1,243 g; FM = 1,053 g; TBW = 881 ml). However, more research on the effects of standardized diet and physical activity on biological precision error is warranted before we can advocate for the integration of pretest diet and physical activity standardization into current best practice methods.

In this study, biological factors, including diet and exercise, contributed a relatively small amount to total error, indicating that measurement error originated primarily from technical sources. There were no significant differences in within- and between-day precision error of FFM, FM, and TBW and the 95% CIs for the differences in within- and between-day precision of FFM and TBW estimates were less than the smallest effect size of interest (0.365% and 0.437%, respectively). This demonstrates that between-day estimates of body composition, particularly FFM and TBW, were very precise.

Some degree of biological error does appear to be inevitable when measuring body composition using BIA, regardless of pretest standardization. Even with strict adherence to best practice and the additional standardization of 24-hr diet and physical activity, the increase in %CV for between-day FM measurements was greater than the smallest effect size of interest of 1%, showing that biological factors impacted the precision of FM measures to at least some extent. There is currently no acceptable level recommended for interrater reliability of BIA (Rose et al., 2021). However, the International Society for Clinical Densitometry (ISCD) has set acceptable limits for dual-energy X-ray absorptiometry (DXA) at <3% when measuring FM, and this threshold has previously been applied to other techniques of measuring body composition (Rose et al., 2021). While the CIs for the mean difference in %CV are wider for FM than for FFM and TBW measures, the upper 95% CI (2.143%) was still within the acceptable limit of 3%, indicating that the technique was able to produce relatively precise between-day measurements, and demonstrated

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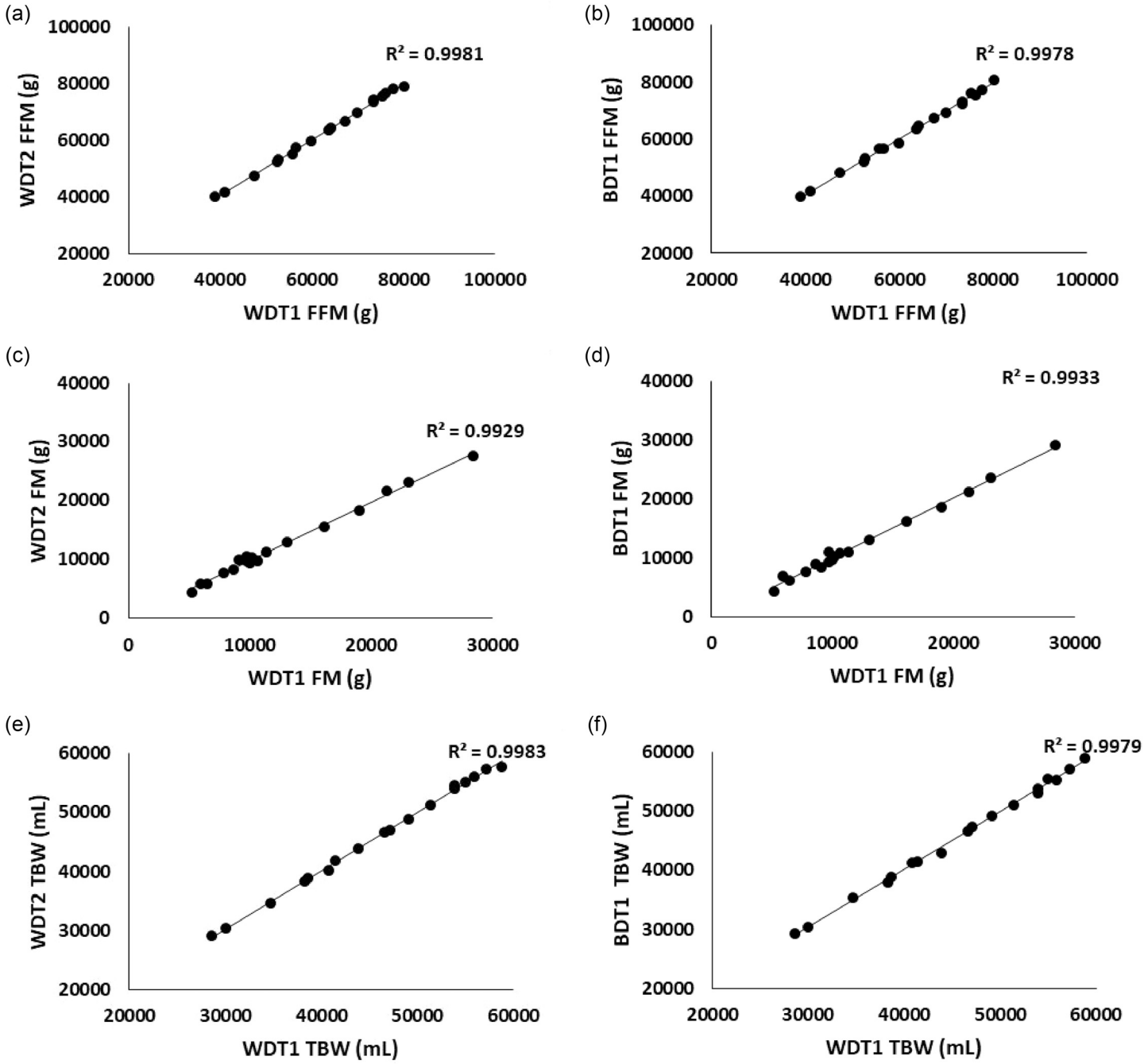

Figure 1 — Regression plots with Pearson's correlations for (a) within-day FFM %CV, (b) between-day FFM %CV, (c) within-day FM %CV, (d) between-day FM %CV, (e) within-day TBW %CV, and (f) between-day TBW %CV. FFM = fat-free mass; FM = fat mass; TBW = total body water; % CV = percentage coefficient of variation; WDT1 = within-day test 1; WDT2 = within-day test 2; BDT1 = between-day test 1.

interrater reliability comparable to other well-accepted methods of measuring body composition. This suggests that with appropriate levels of pretest standardization, impedance may have application in the longitudinal tracking of athletic populations.

In practice, minimizing LSC is highly important, as athletes tend to experience comparably small changes in body composition over a given length of time and therefore require highly precise assessment tools (Moon, 2013). In this study, the LSC values for body composition measures were markedly smaller than has been previously reported in the literature. Previous research by Farley et al. (2020), who measured within- and between-day precision of FM and FFM estimates with Impedimed SFB7 Bioelectrical Impedance Spectroscopy (BIS), found error to be higher compared to other techniques such as DXA and air displacement plethysmography. Due to the high LSC values reported for within-day (FFM = 2,276 g, FM = 2,331 g) and between-day (FFM = 3,966 g, FM = 3,607 g) estimates of body composition, it was concluded that it would be difficult to track physique changes in athletes using BIS. The replication of dietary intake and physical activity in the 24 hr prior to testing is the most notable difference in the two methodologies and likely contributed to

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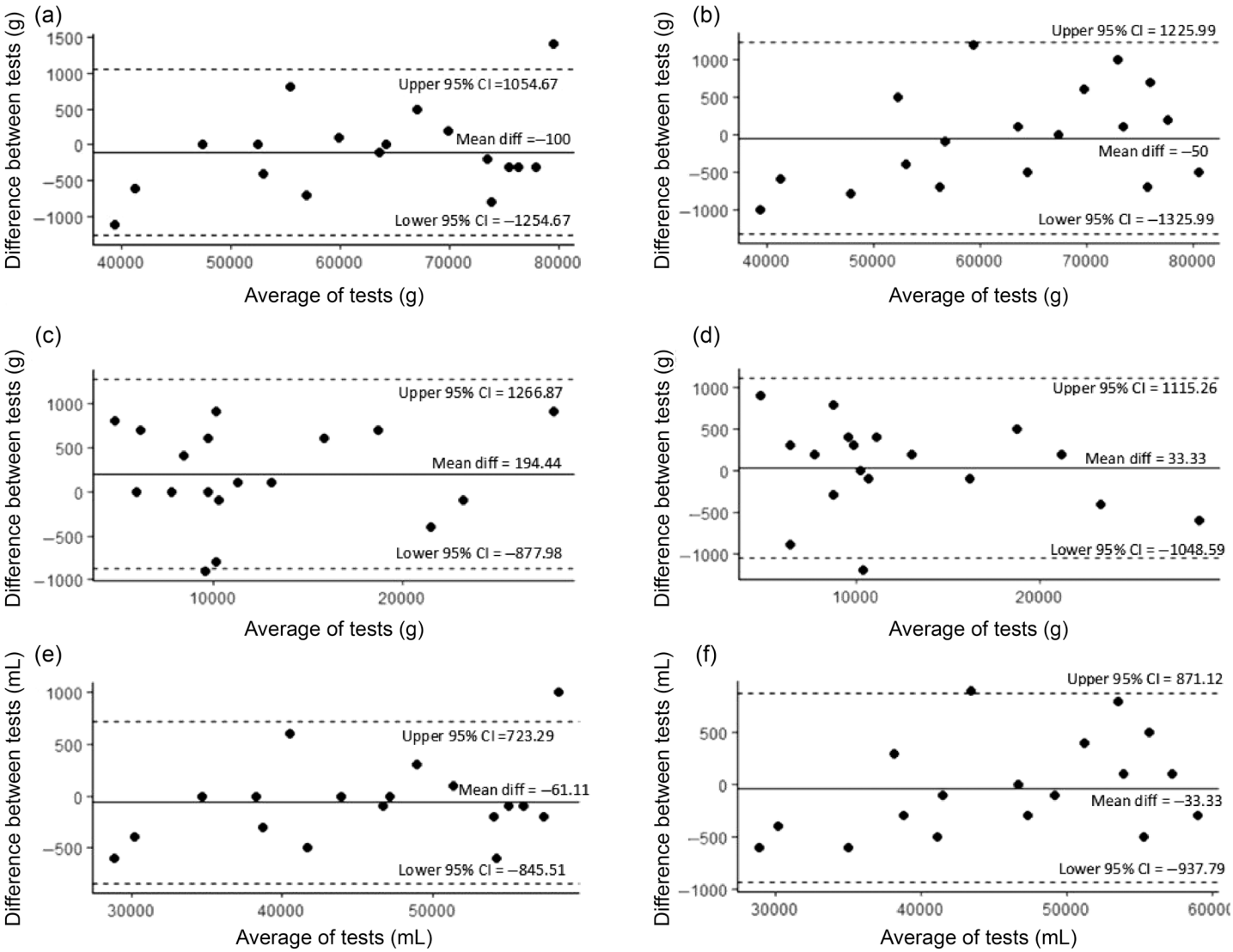

Figure 2 — Bland–Altman plots with 95% limits of agreement of raw measurements (in grams or milliliters) for (a) within-day FFM, (b) between-day FFM, (c) within-day FM, (d) between-day FM, (e) within-day TBW, and (f) between-day TBW. FFM = fat-free mass; FM = fat mass; TBW = total body water; CI = confidence interval; Mean diff = mean difference.

Table 2 Summary of Within-Day (WDT1 and WDT2) Precision Error for FFM, FM, and TBW

Body composition measure	WDT1 (g; mean ± *SD*)	WDT2 (g; Mean ± *SD*)	Within-day difference (g; mean ± *SD*)	RMS_*SD*	LSC (g)	%CV (mean ± *SD*)	*p*	R^2
FFM	62,567 ± 12,706	62,667 ± 12,503	−100 ± 589	411	1,138	0.52 ± 0.53	.48	.9981
FM	12,533 ± 6,472.29	12,339 ± 6,432	194 ± 547	400	1,109	3.18 ± 3.34	.15	.9929
TBW	45,844 ± 9,350	45,906 ± 9,220	−61 ± 400	278	771	0.46 ± 0.47	.53	.9983

Note. FFM = fat-free mass; FM = fat mass; TBW = total body water; RMS_*SD* = root mean square standard deviation; LSC = least significant change; %CV = percentage coefficient of variation; WDT = within-day test.

the improved precision observed within this study. However, several other factors may have precipitated these discrepancies in precision. For example, nuances in the subject population may have impacted precision error. In particular, Farley et al. (2020) used a population of large, muscular males, as opposed to this study which used mixed males and females from a variety of different sports. These differences in participant characteristics have been shown to impact the precision of body composition estimates (Curtis et al., 2022). Differences in the technology may also have impacted the precision of body composition data (Ward, 2019).

Physique differences across different population groups have been shown to impact the applicability of BIA equations for

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Table 3 Summary of Between-Day (WDT1 and BDT1) Precision Error for FFM, FM, and TBW

Body composition measure	WDT1 (g; mean ± *SD*)	BDT1 (g; mean ± *SD*)	Between-day difference (g; mean ± *SD*)	RMS_*SD*	LSC (g)	%CV (mean ± *SD*)	*p*	R^2
FFM	62,567 ± 12,706	62,617 ± 12,441	−50 ± 651	449	1,243	0.66 ± 0.49	.75	0.9978
FM	12,533 ± 6,472	12,500 ± 6,610	33 ± 552	380	1,053	3.38 ± 3.71	.80	0.9933
TBW	45,844.44 ± 9,350	45,878 ± 9,165	−33 ± 461	318	881	0.63 ± 0.46	.76	0.9979

Note. FFM = fat-free mass; FM = fat mass; TBW = total body water; RMS_*SD* = root mean square standard deviation; LSC = least significant change; %CV = percentage coefficient of variation; WDT = within-day test; BDT = between-day test.

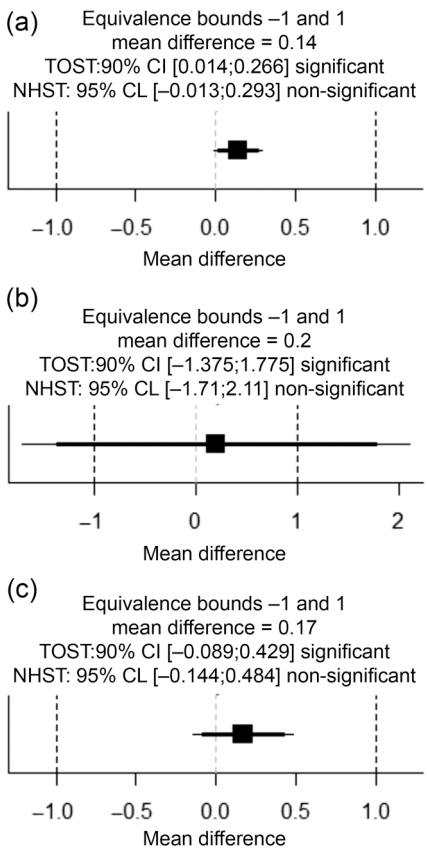

Figure 3 — Equivalence plots showing the mean difference in %CV and 95% confidence intervals for (a) FFM, (b) FM, and (c) TBW, with the upper equivalence bound set at 1% to correspond to an acceptable level of interrater reliability for anthropometric techniques. FFM = fat-free mass; FM = fat mass; TBW = total body water; %CV = percentage coefficient of variation; CL = confidence limit; TOST = two one-sided T-tests; NHST = null hypothesis significance testing.

estimating body composition. The BIA technique assumes that the limbs and trunk are cylindrical conductors with a typical density (Kyle et al., 2004b); however, athletes with high muscularity have atypical density due to the fluid content of the FFM (Kerr et al., 2015), which may alter the validity of predictive equations. Few BIA equations have been validated for use in specific athletic populations (Meyer et al., 2013). In a previous study conducted in a population of resistance-trained males, over half of the participants had a TBW that differed >1% from the assumed value of 73.72% hydration of the FFM, which introduced error into percentage body fat (%BF) estimations compared to reference Model 3C (Kerr et al., 2015). Previous research also indicates proportional bias when measuring %BF in athletes using foot-to-foot BIA, with overestimation when there is a low %BF and vice versa (Nickerson et al., 2019).

Intermachine variability has also been cited as a key factor contributing to the ongoing criticism toward the validity of the BIA technique (Ward, 2019). The Impedimed model used by Farley et al. (2020) is a BIS model that uses Cole–Cole modeling to estimate resistance at 0 (representing extracellular water [ECW] only) and infinite (representing TBW or the sum of ECW + intracellular water) frequencies by extrapolation (Kyle et al., 2004b). While this approach may be more accurate in theory, it does not always translate into practice (Yamada et al., 2013), possibly due to a reliance on extrapolation. Other factors that have been identified as contributing to error in BIS are stray capacitance, where manifestation of charge in noncapacitive components occurs, impacting measurements at a high frequency, and electrode polarization where the presence of ions at the interface impacts the measurement precision, impacting measurements at a low frequency (Naranjo-Hernandez, Reina-Tosina, & Min, 2019; Naranjo-Hernandez, Reina-Tosina, Roa, et al., 2019). Furthermore, the use of manually attached electrodes may introduce additional sources of error, such as issues of contact between the electrode and skin due to imperfect attachment or the condition of the skin (Naranjo-Hernandez, Reina-Tosina, & Min, 2019; Naranjo-Hernandez, Reina-Tosina, Roa, et al., 2019).

In the past, InBody BIA has been shown to have good agreement with DXA, with strong correlations (Lahav et al., 2021) and low individual error (McLester et al., 2020), though both systematic (underestimation of %BF) and proportional biases (deviation dependent on body mass index group or gender) were present. By comparison, a SECA multifrequency bioelectrical impedance analysis (MF-BIA) model did not underestimate %BF and was unaffected by body mass index range and gender (Lahav et al., 2021). However, clinically significant individual differences have been found to occur between InBody (MF-BIA), Xitron (BIS), and Tanita (single frequency bioelectrical impedance analysis [SF-BIA]) models, suggesting that variance in body composition may occur due to methodological differences in posture (orthostatic vs. supine), differences in electrode configuration (tetrapolar vs. eight attachment points), or materials (disposable electrodes with conducive gel vs. tactile electrodes; Guedes et al., 2019).

While many studies assess the accuracy of different BIA models for measuring the "true" values compared to a criterion, there is less research investigating the precision of various models for longitudinal measures (Tinsley et al., 2021). A study by Tinsley et al. (2021) reported a low technical error of measurement for the InBody 770 model for measuring FM (0.20 kg) and FFM (0.19 kg) as well as low associated %CV. Interestingly, the same Impedimed

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system used by Farley et al. (2020; SBF7) was reported by Tinsley et al. as having lower technical error of measurement than the InBody device. However, the InBody had better agreement with a 4C model than Impedimed when measuring changes in body composition after 6 weeks (Tinsley et al., 2021), suggesting that the SBF7 may be more sensitive to differences of subject presentation. This example highlights a key reason why future investigations should focus on the ability of various models to track body composition changes over time, as opposed to their agreement with other "criterion" techniques. While systematic biases are likely an inevitable component of the BIA technique (Ward, 2019), more clinically significant is the ability of various models to reproduce precise values for repeated measurements. Accordingly, his should drive the future research in this field.

This study has several limitations. First, the study design does not include a day of randomized dietary intake and physical activity as a control, which does not allow conclusions to be drawn as to the effectiveness of the 24-hr standardization protocol compared to the current best practice. Second, the sample size is smaller than the ISCD recommended number of 30 participants for calculating LSC, which has implications for the statistical power of the sample. Additionally, as food was not provided to participants, the standardization of diet relied on participants replicating their own diet, which may have reduced adherence to this protocol. The diet and training diary also did not account for the impact of incidental activity, which may have varied day-to-day. Although attempts were made to standardize subject presentation, menstrual status and postural shifts were not accounted for. While it appears phase of menstrual cycle has minimal impact on estimates of body composition via BIA (Rael et al., 2021), we acknowledge standardization of posture should be advocated in future studies (Gibson et al., 2015).

The current best practice guidelines advocate presenting overnight fasted and rested states to minimize the impact of dietary intake and physical activity on BIA precision; however, this approach is likely insufficient to ensure stabilization of hydration status and acute shifts in solutes between the intracellular fluid and extracellular fluid. This study demonstrates that the 24-hr standardization of diet and physical activity appears to be a promising approach to minimizing biological error when estimating body composition using BIA. In practice, sports science practitioners could adopt this protocol using a diet and training diary to standardize athlete diet and physical activity in the 24 hr prior to testing. This may afford more precise estimates of change in body composition. However, further research is needed before advocating for inclusion of this protocol in future best practice guidelines. Such research should seek to confirm the validity of a 24-hr standardization protocol for minimizing precision error of impedance estimates compared to a 24-hr randomized or ad libitum dietary intake and physical activity. Furthermore, it would be worthwhile comparing the effect of this protocol in different MF-BIA and BIS systems, given that the results of this study suggest there may be marked differences between models.

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